How the PCRMs site works

A map of the code, for reading it or changing it. It describes the site as of October 2026: the K-green redesign, the third-normal-form database and the human and mouse data.

PCRMs
Predicted cis-Regulatory Modules

1 Pages are PHP files

Each address is one file in `www/`, such as `crm.php`. Shared code lives in `includes/`, outside the public folder, so it can't be opened from the web.

2 Searches are saved, then shown

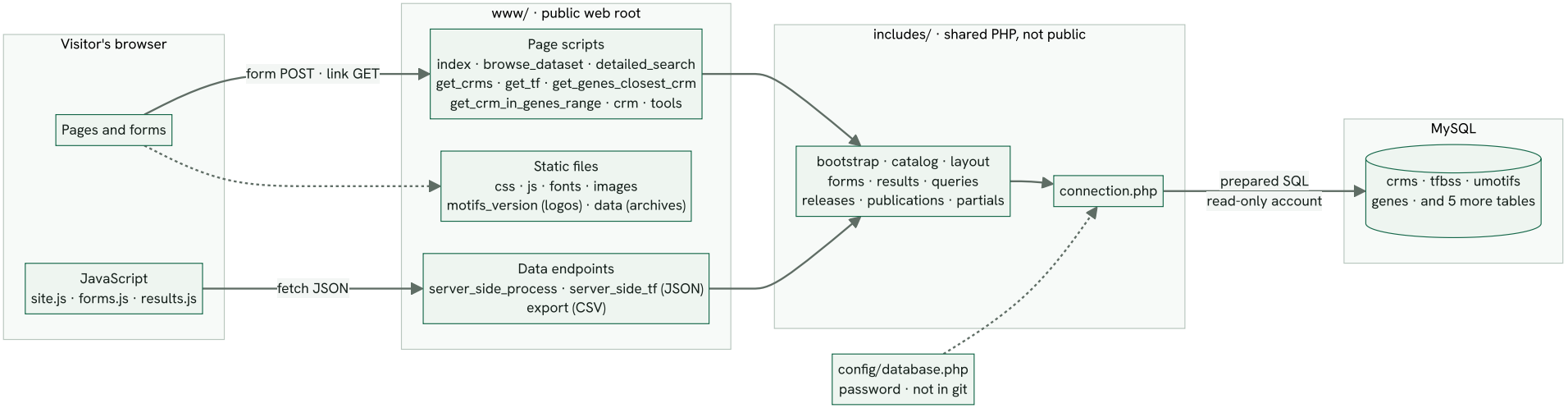
A search form is checked, saved on the server under a random token, and the browser is sent to `…?query=TOKEN`. The results can be reloaded and paged in that browser for 6 hours.

3 Tables work without JavaScript

Page, sort and filter live in the address. PHP draws the first page; `js/results.js` then redraws the table in place as visitors click.

4 The database is read-only

The site only reads MySQL. The data is loaded offline from your files by the scripts in `database/`.



Where things are

Only `www/` is visible on the web. Everything else sits beside it on the server, or stays on your computer.

```
www/                  public: the web server's document root
├─ *.php              one file per page or data endpoint (page 3)
├─ js/                site.js · forms.js · results.js
├─ css/modern.css     all styles; colours at the top
├─ fonts/ images/     Hanken Grotesk · logos · favicon
├─ motifs_version/    logos: 1/ human, 2/ mouse (SVG), 3/ worm (PNG)
├─ data/              download files (.bed.gz), not in git
└─ .htaccess          404 page, no folder listings
includes/             shared PHP, loaded by the pages (page 5)
├─ partials/          site header (menu) and footer
└─ svg/               organism and pipeline icons, inlined
config/               database.php: login details, not in git
database/             schema, loaders, data scripts (page 8)
data-incoming/        your raw data: not in git
docs/                 this guide
README · DEPLOYMENT · DATA_NEEDED  notes (.md)
```



Words the code uses

CRM	cis-regulatory module: a region with a score and P-value (table `crms`).
TFBS	a binding site inside a CRM (table `tfbss`).
umotif / umID	a unique motif from ProSampler and its number within a map, as in `328.svg`.
dataset	one CRM map of a genome: dePCRM2 now, Nucleotide Transformer later.
genome ID	1 human, 2 mouse, 3 worm, 4 thale cress, 5 fruit fly.
token	the random 32-character name a saved search gets.
remote / local table	a results table that pages from the server, or one that has every row in the page.
`pcrms_` prefix	every shared PHP function starts with it, so it is easy to search for.

> To run it locally: `php -S 127.0.0.1:8765 -t www` with a database set in `config/database.php`. The steps are in `README.md`.

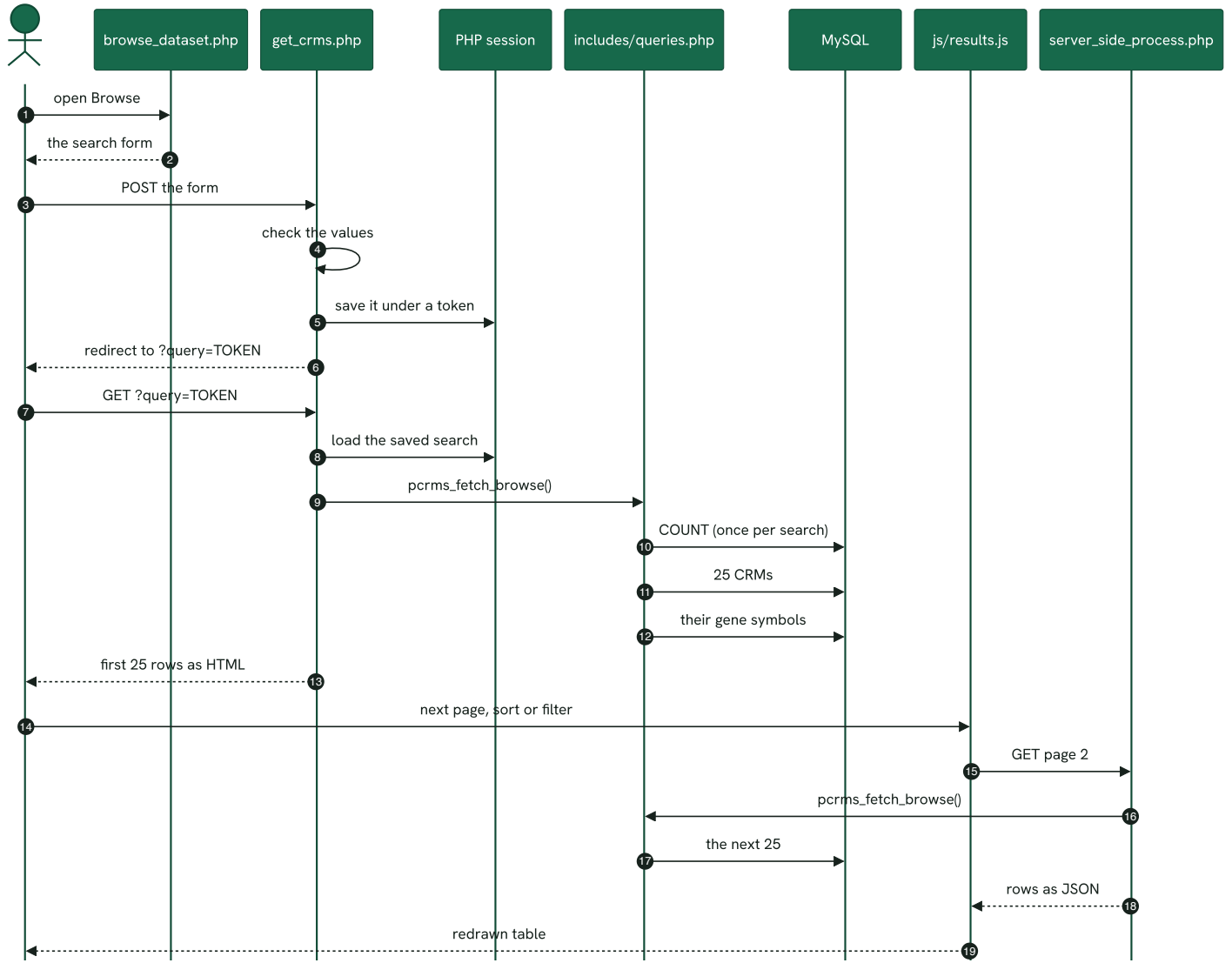
Every page and what it reads

The menu leads to four pages; the others are results, records, data endpoints and old addresses kept alive.

FILE IN WWW/	WHAT VISITORS SEE OR GET	MAIN CODE IT USES	TABLES IT READS
MENU PAGES			
index.php	Home: CRM counts per genome, how the predictions are made, publications	layout · releases.php · publications.php	none (counts are typed into releases.php)
browse_dataset.php	Browse: tabs for dePCRM2 and Comparative maps; genome, chromosome and cutoff form	forms.php	none
detailed_search.php	Functional analysis: three search tabs (closest gene, around a gene, TF)	forms.php · forms.js	none
tools.php · download_data.php	Tools (ProSampler, coming soon) · release archives from `www/data/`	layout · releases.php	none
SEARCH RESULTS			
get_crms.php	Browse results; remote table	queries.php `pcrms_fetch_browse()`	crms · crms_genes_association · genes
get_tf.php	Binding sites of a TF; remote table	queries.php `pcrms_fetch_tf()`	umotif_tfs_association · umotifs · tfbss · crms
get_genes_closest_crm.php	CRMs whose closest gene is the one entered; local table	SQL in the page	genes · crms_genes_association · crms
get_crm_in_genes_range.php	CRMs within 0.5, 1 or 2 Mb of a gene; local table	SQL in the page	genes · crms
crm.php	One CRM: summary and its binding sites. Up to 1,000 sites in the page; larger CRMs page from `crm.php?format=json`	results.php · queries.php	crms · tfbss · umotifs · TF-name view · genes
DATA ENDPOINTS (NO PAGE OF THEIR OWN)			
server_side_process.php	JSON rows for the browse table, one page at a time	queries.php	as get_crms.php
server_side_tf.php	JSON rows for the TF table	queries.php	as get_tf.php
export.php	CSV of a browse or TF search, up to 250,000 rows	queries.php	as the search
OLD ADDRESSES, KEPT WORKING			
tfbs.php · DS_*.php · get_genes.php	Permanent redirects to crm.php and detailed_search.php?tool=…	catalog.php	none
404.php	“Page not found”, set in `www/.htaccess`	layout	none

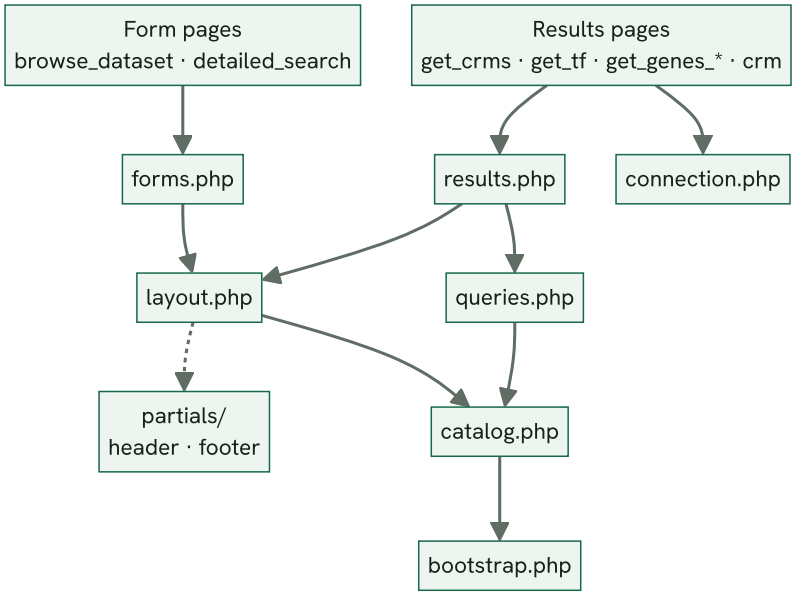
One search, step by step

A browse search, from the form to page 2 of the results. The TF search follows the same path through `get_tf.php` and `server_side_tf.php`. The count runs once per search and is then kept in the session; the first page is plain HTML, so it works without JavaScript.



The shared code in includes/

Pages load what they need with `require_once`; each file loads the ones it depends on (arrows), so `bootstrap.php` always runs first.

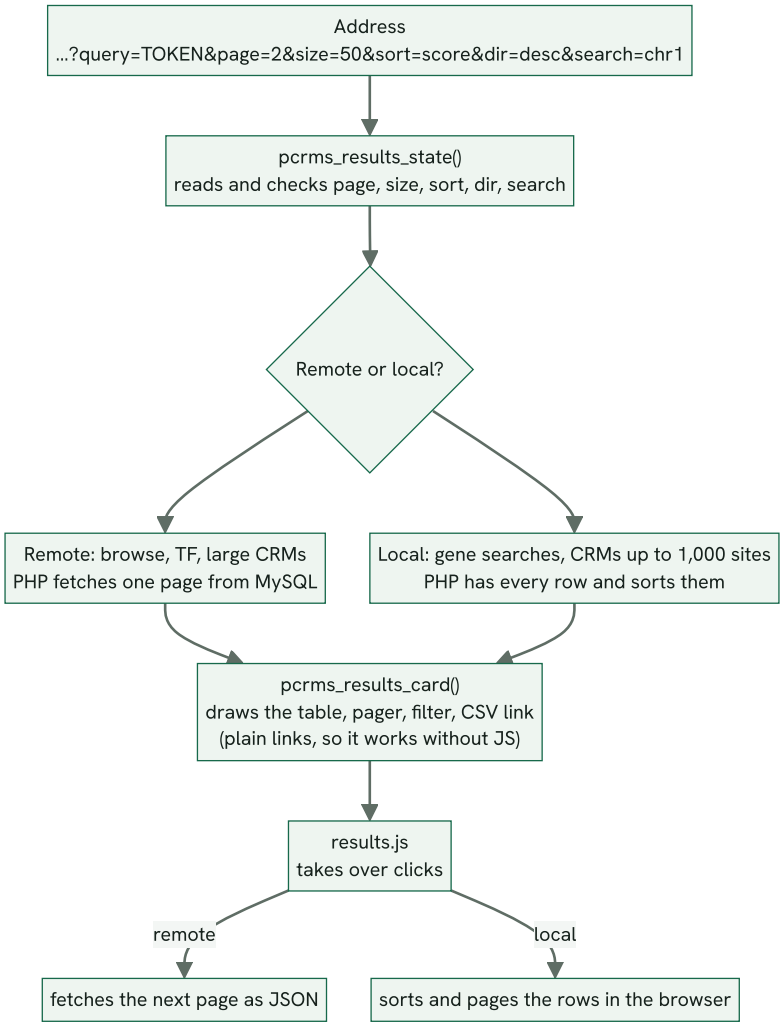


FILE	WHAT IT HOLDS
bootstrap.php	Security headers on every response, PHP errors hidden from visitors, session cookie settings (`pcrms_session_start()`).
catalog.php	The genome list `$PCRMS_GENOMES` (names, assemblies, chromosomes, UCSC names, `searchable`), the cutoffs `$PCRMS_P_VALUES`, checks `pcrms_validate_*()`, `pcrms_h()` for safe HTML, saved searches and their counts, error pages, UCSC links, `pcrms_execute_statement()`.
layout.php	`pcrms_page_start()` and `pcrms_page_end()` wrap every page: head, menu, footer, scripts. `pcrms_svg()` inlines icons.
forms.php	The genome cards, chromosome chips, cutoff pills and search button used by both search pages.
results.php	The results table: its columns (`$PCRMS_RESULT_COLUMNS`), reading page, sort and filter from the address, drawing the table, sorting small tables in PHP, CSV output.
queries.php	SQL for the browse and TF searches: building WHERE and ORDER BY, counting, fetching one page, turning rows into table rows. Also the CSV row limit.
connection.php	Connects with `config/database.php` or `PCRMS_DB_*` settings and limits each query to 15 seconds.
releases.php · publications.php	Download list and home-page counts · citation and related papers.

JavaScript: `site.js` opens the phone menu and the “more papers” list; `forms.js` updates the search summary, the chromosome buttons and the analysis tabs; `results.js` redraws result tables and copies the PHP formatting functions so both look the same.

How a results table works

All four tables (browse, TF, gene searches, CRM record) come from one function, `pcrms_results_card()`, and one script, `results.js`.



The four tables

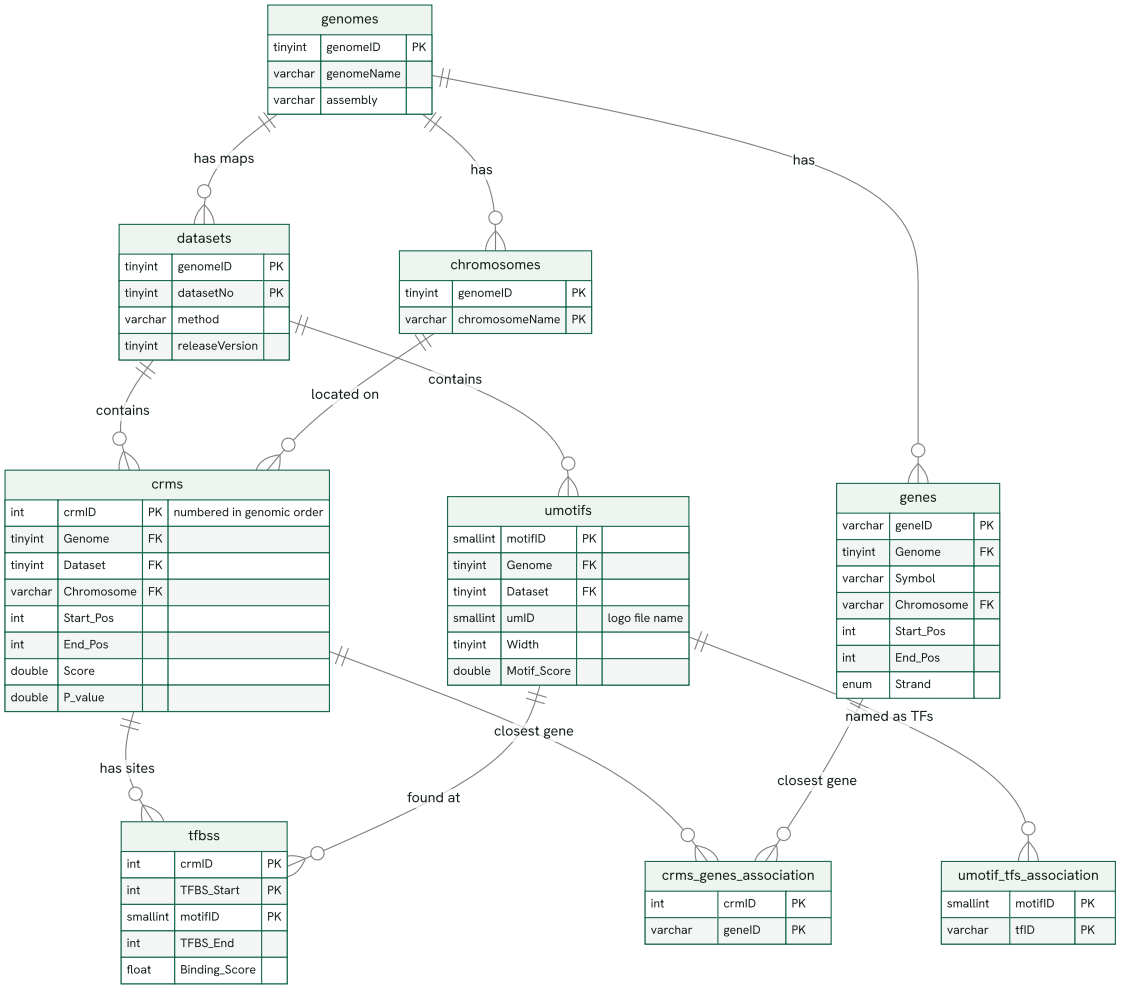
TABLE KEY	USED BY	MODE
crms	get_crms.php	remote · server_side_process.php
tf-sites	get_tf.php	remote · server_side_tf.php
gene-crms	both gene searches	local
crm-sites	crm.php	local, or remote above 1,000 sites

Good to know

- Columns are listed once, in `$PCRMS_RESULT_COLUMNS`; each has a type (location, score, p, logo…) that decides how it is drawn, sorted and searched.
- Only non-default settings go into the address, so links stay short.
- Counting a large search is the slow part, so the counts are saved with the search and reused while paging.
- Sorting tens of millions of binding sites by score can pass the 15-second limit; the table then says the search took too long.
- CSV downloads stream from `export.php` (browse and TF) or come from the page itself with `&format=csv`.

The database

Defined in `database/schema.sql`, in third normal form: each fact is stored once. Human and mouse hold 2.3 million CRMs and 290 million binding sites, about 16 GB.



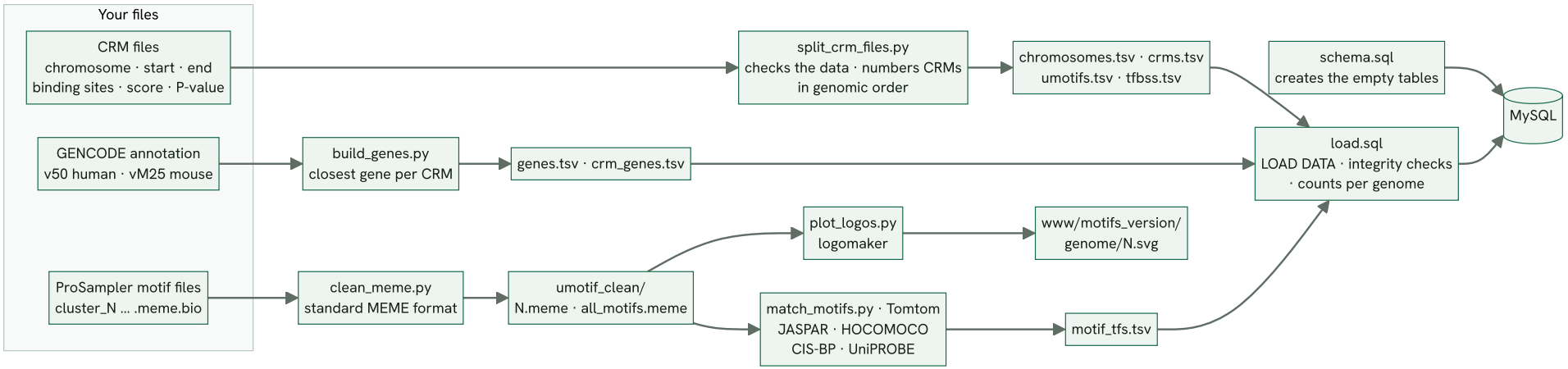
Reading it

- **A dataset** is one CRM map of one genome. Human and mouse have dataset 1 (dePCRM2 v2); a Nucleotide Transformer map would be dataset 2.
- **CRM IDs follow the genome**: chr1 first, by start. So one chromosome's CRMs are one block of IDs, and the TF search reads only that block of binding sites.
- **Motif facts sit in `umotifs`**, once per motif, not on each of 290 million sites.
- **Lists are built when asked**: gene symbols per CRM by the queries, TF names per motif by the one view, `umotif_tfs_association_multiple`.
- **TF names** in `umotif_tfs_association` come from matching the motifs against JASPAR, HOCOMOCO, CIS-BP and UniPROBE (page 8).
- **Genes** are GENCODE's protein-coding genes; each CRM's closest gene is the one whose start site is nearest (page 8).

> The site's account can only read these tables: `database/grants.sql`.

Getting data in

Done on your computer or the server, not by the website. Each step checks the data and stops with a clear message if something is wrong. Details are in `database/README.md`.



- Splitting human and mouse takes about 10 minutes; loading took 28 minutes on a laptop.
- Load with `mysql --local-infile=1 --show-warnings`: any warning means rows were skipped.
- To switch on fly, worm or thale cress: load their files, then set `'searchable' => true` in `includes/catalog.php`.
- `synthetic_fixture.sql` fills the same tables with small made-up data for testing.

Safety built in

What protects the site and the database, and where each protection lives. `DEPLOYMENT.md` lists what the server admin still has to set up.

PROTECTION	WHAT IT PREVENTS	WHERE
Every input checked against allowed values	odd genome IDs, chromosomes, cutoffs or IDs reaching the database	catalog.php `pcrms_validate_*()`
Placeholders in all SQL	SQL injection: values never become part of the SQL text	`pcrms_execute_statement()`
Output escaped	cross-site scripting through data or the address	`pcrms_h()`; JSON with escaping flags
Content-Security-Policy and other headers	injected scripts, the site being framed by others	bootstrap.php
Code outside the public folder	visitors opening include or config files	includes/ · config/ · www/.htaccess
Read-only database account	any change to the data through the website	database/grants.sql
15-second limit per query (120 s for CSV)	one heavy search tying up the server	connection.php
Searches saved under random tokens for 6 hours	guessing or tampering with someone else's search	catalog.php `pcrms_store_query_state()`
Session cookie: HttpOnly, SameSite, Secure on HTTPS	scripts or other sites reading the cookie	bootstrap.php
CSV cells starting with = + - @ made inert	spreadsheet formulas running from a download	queries.php `pcrms_csv_cell()`
Errors logged, not shown	server details leaking in error messages	bootstrap.php · `pcrms_service_unavailable()`

> Still for the server admin (DEPLOYMENT.md): new database passwords, the read-only account, HTTPS, rate limits on the search pages, and removing the old site's leftover files.

Where to change things

The usual edits, and the file to open. After any change, try one search of each kind on the local site.

TO CHANGE…	EDIT
Home page text	www/index.php
CRM counts and download list	includes/releases.php (files in www/data/)
Citation and related papers	includes/publications.php
Menu	includes/partials/site_header.php
Footer	includes/partials/site_footer.php
Colours, spacing, fonts	www/css/modern.css (colour list at the top)
Tools page	www/tools.php
Browse tabs, Comparative maps	www/browse_dataset.php (`$mapTabs`)

TO CHANGE…	EDIT
P-value cutoffs	includes/catalog.php `$PCRMS_P_VALUES`
Switch a genome on or off	catalog.php `'searchable'`
Add a genome	catalog.php `$PCRMS_GENOMES` and `$PCRMS_GENOME_ORDER` , an icon in includes/svg/organisms/, rows in database/load.sql
Results table columns	results.php `$PCRMS_RESULT_COLUMNS` , plus the row builders in queries.php
A search's SQL	queries.php (browse, TF) or the get_genes_*.php page
Database login	config/database.php (never in git)
Load new data or redraw logos	database/README.md
Security headers	includes/bootstrap.php

> If you change the one inline script in `layout.php` , update its hash in `bootstrap.php` , or browsers will block it; the command is in the comment there.